

CCT College Dublin

Assessment Cover Page

Module Title:	BSc in Computing and IT
Assessment Title:	CA1 – Virtual Windows Active Directory / Network Server Virtualization
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Operating Systems

CA1 – Virtual Windows Active Directory / Network Server Virtualization

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Introduction

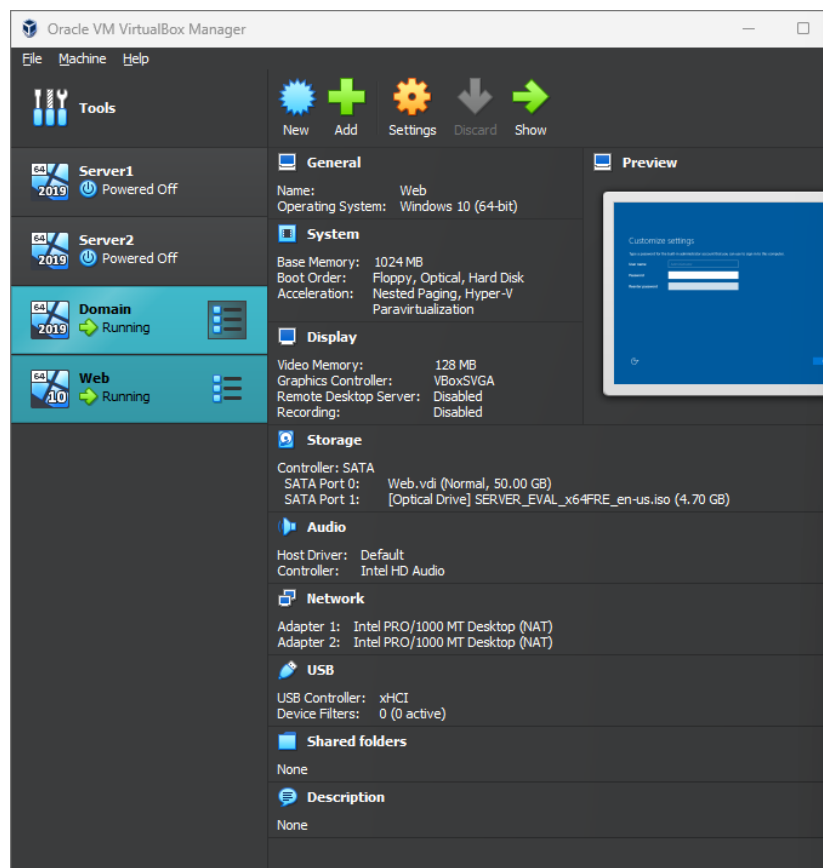
In this first four parts of this assignment, I'll be setting up two new virtual machines for the DigiTech company. The first part is creating and configuring the machines, both the Domain Server and Web Server, accordingly to the required specifications given: from the new machine's hostnames, to assigning the IP and DNS addresses, testing and connecting, setting the active domain and adding new users on the organizational units and setting passwords and permissions for them, to have a fully working Domain and Web Servers for the company.

The fifth part of the assignment I'll show my research and a case study on the importance of patch management for the networking servers and explain about Windows Server Update Services (WSUS). At the end, in part six, I'll be doing some challenges about Group Policy Management, like sharing folders, installing software on servers, adding a wallpaper and restricting the Panel Control.

Specific Requirements for this virtualized network:

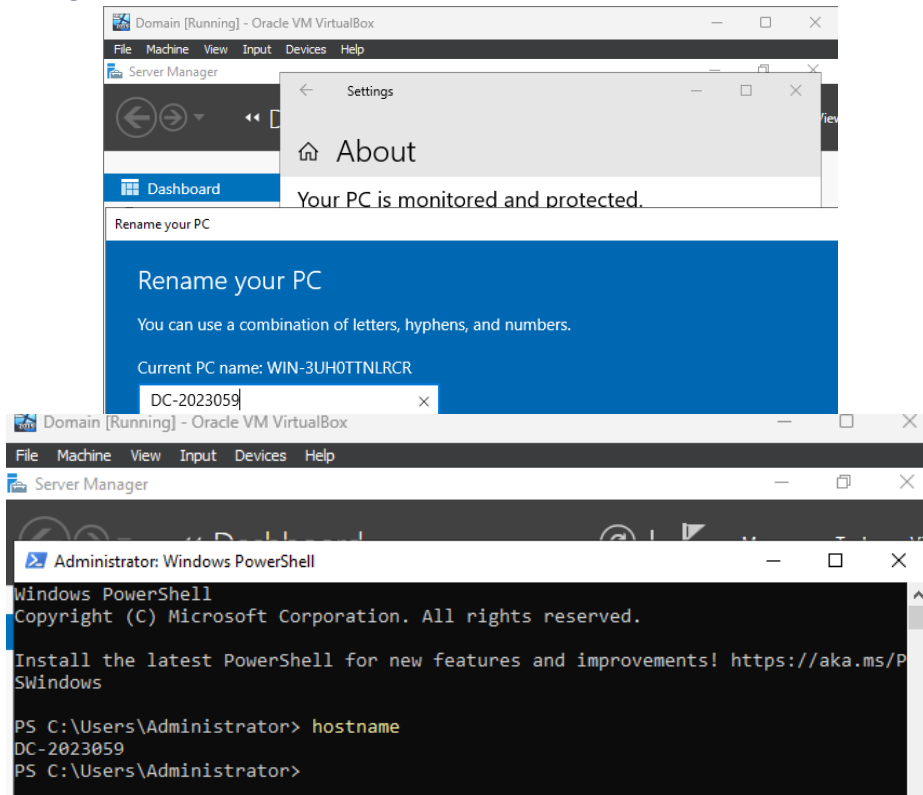
Part 1: Installing the network and setting up the network infrastructure.

Creating two new Servers for the DigiTech, one to be as the Domain server and the other to be the Web server.



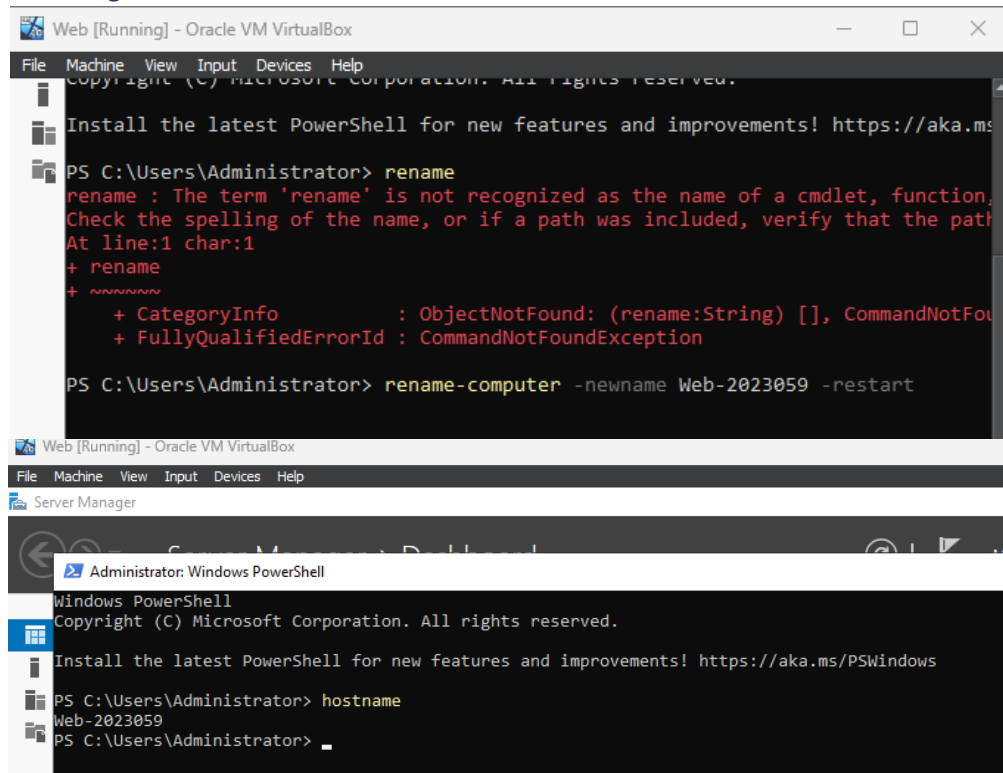
1 The two new servers on Oracle VM Virtual Box

1.A Renaming the Domain Server



2 Renaming using the Windows System Properties – DC-2023059

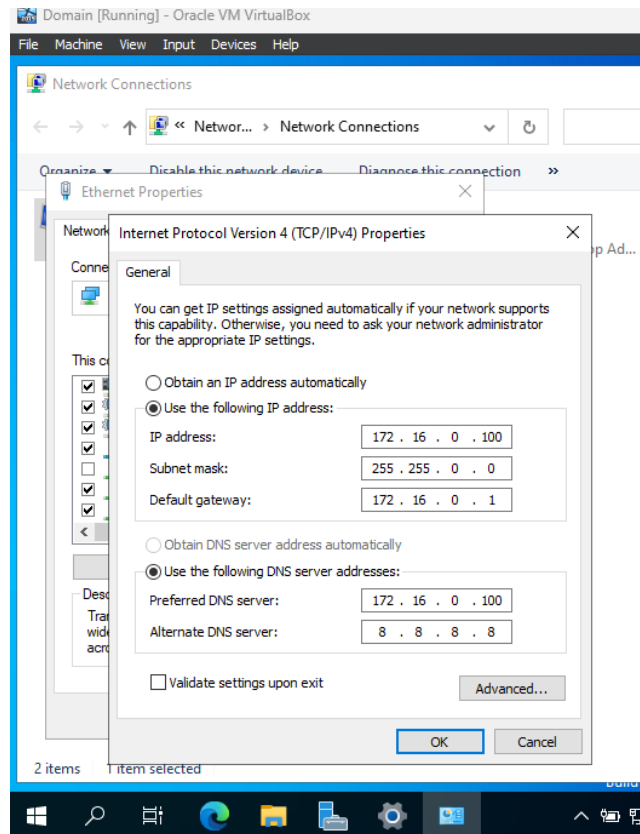
1.A Renaming Web Server



3 Renaming using the PowerShell - Web-2023059

1.B Assigning the IP addresses in both machines

Domain Server



4 Assigning the IP Address in the Domain Server

On the Domain Server I set the following IP Address and DNS:

IP address: 172.16.0.100

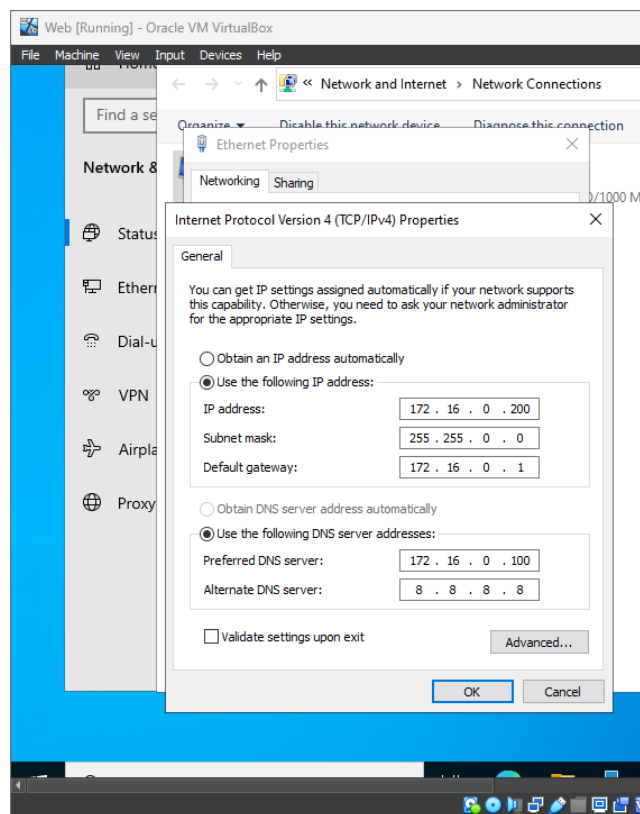
Subnet mask: 255.255.0.0

Default gateway: 172.16.0.1

DNS server: 172.16.0.100

Alternate DNS server: 8.8.8.8

Web Server



5 Assigning the IP Address in the Web Server

On the Web Server I set the following IP Address and DNS:

IP address: 172.16.0.200

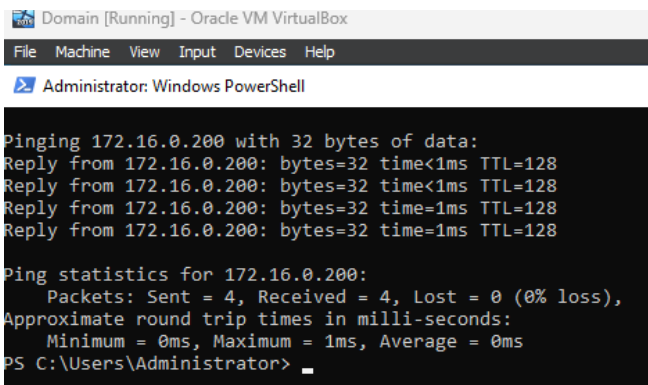
Subnet mask: 255.255.0.0

Default gateway: 172.16.0.1

DNS server: 172.16.0.100

Alternate DNS server: 8.8.8.8

1.B Test and Demonstrate Connectivity



```

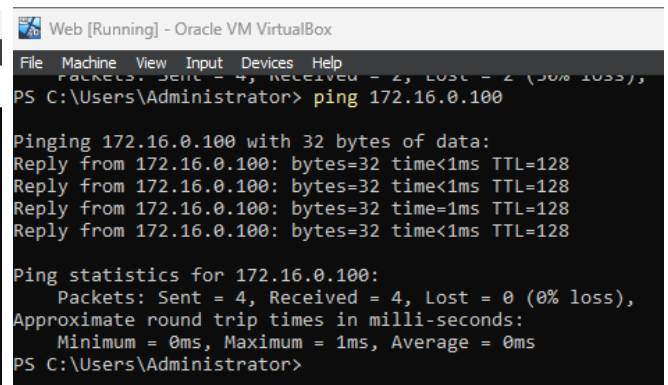
Domain [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Administrator: Windows PowerShell

Pinging 172.16.0.200 with 32 bytes of data:
Reply from 172.16.0.200: bytes=32 time<1ms TTL=128
Reply from 172.16.0.200: bytes=32 time<1ms TTL=128
Reply from 172.16.0.200: bytes=32 time<1ms TTL=128
Reply from 172.16.0.200: bytes=32 time<1ms TTL=128

Ping statistics for 172.16.0.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
PS C:\Users\Administrator>

```

7 Pinging the Web on the Domain



```

Web [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
PS C:\Users\Administrator> ping 172.16.0.100

Pinging 172.16.0.100 with 32 bytes of data:
Reply from 172.16.0.100: bytes=32 time<1ms TTL=128
Reply from 172.16.0.100: bytes=32 time<1ms TTL=128
Reply from 172.16.0.100: bytes=32 time<1ms TTL=128
Reply from 172.16.0.100: bytes=32 time<1ms TTL=128

Ping statistics for 172.16.0.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
PS C:\Users\Administrator>

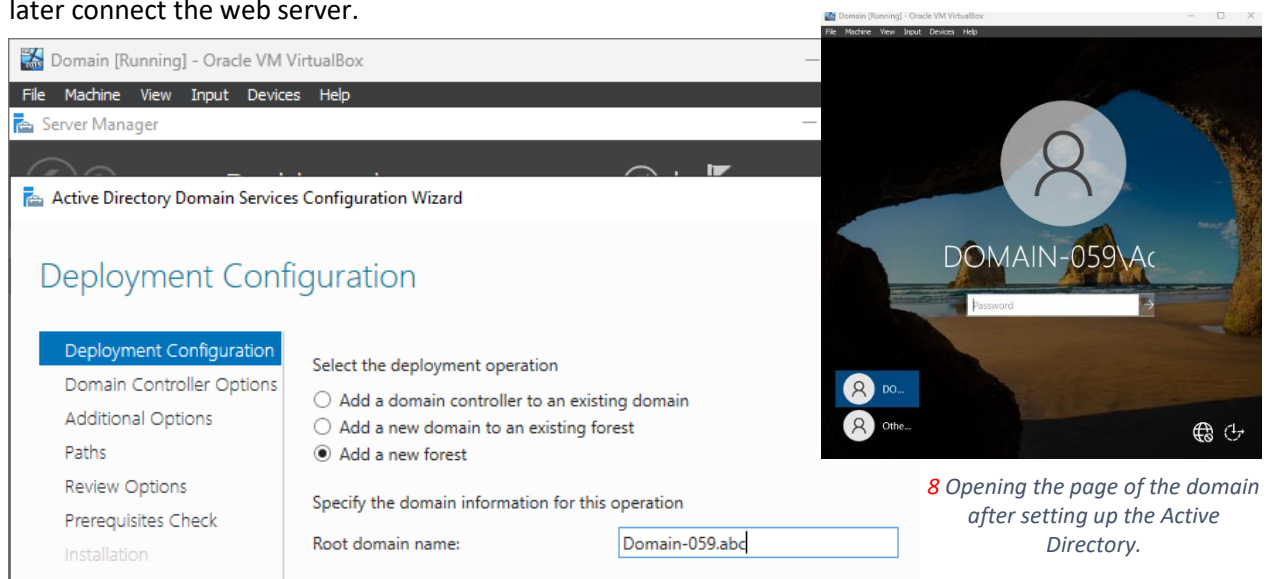
```

6 Pinging the Domain on the Web

Successfully pinging the Web Server (172.16.0.200) on the PowerShell of the Domain (172.16.0.100) (Figure 7) and the pinging the Domain (172.16.0.100) on the PowerShell of the Web Server (172.16.0.200) (Figure 6).

1.C Active Directory Domain

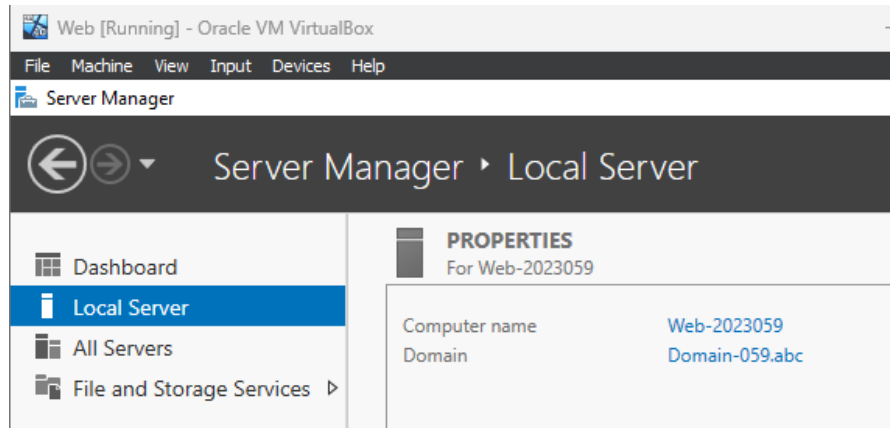
Downloading and setting up the Active Directory Domain on the DigiTech Domain Server to later connect the web server.



9 Adding the root Domain name on the Active Directory Configuration

8 Opening the page of the domain after setting up the Active Directory.

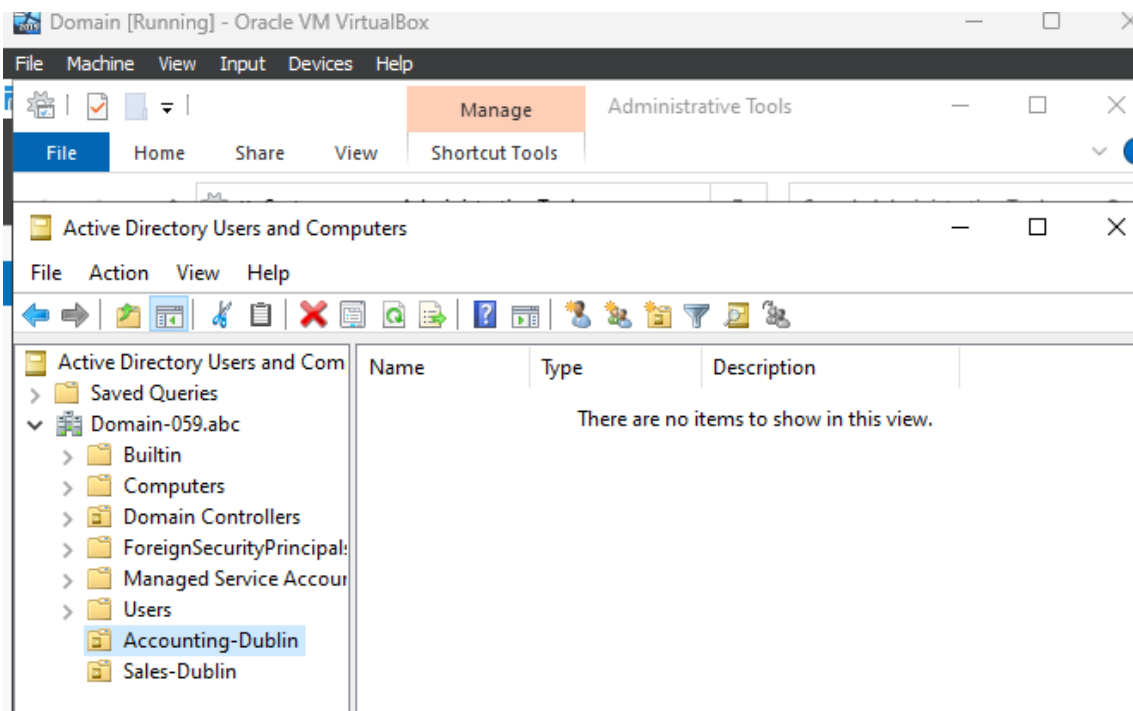
1.D Add the Web to the Domain



10 Web Server connected to the Domain server

1.E Organizational Units

Using the Active Directory User and Computers (ADUC), the DigiTech company added two departments on Organizational Units, Accounting-Dublin and Sales-Dublin.

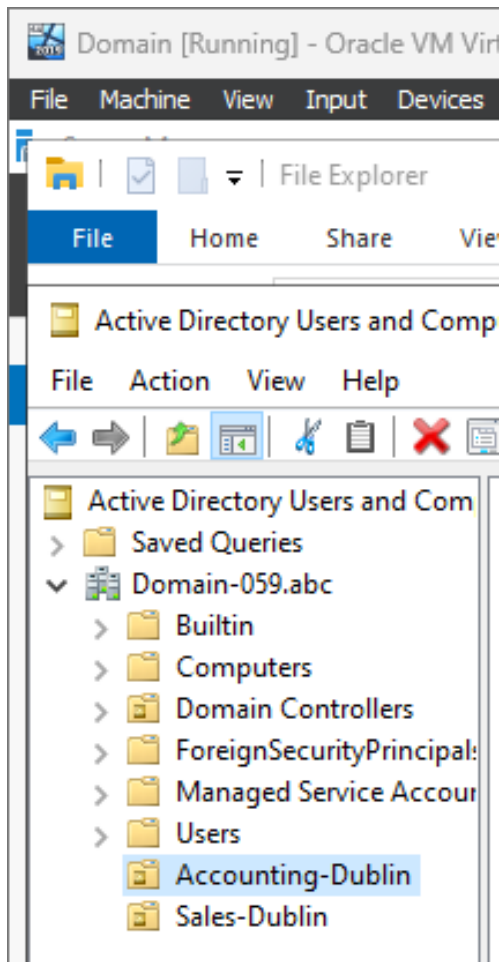


11 The two new departments on ADUC - Accounting-Dublin and Sales-Dublin.

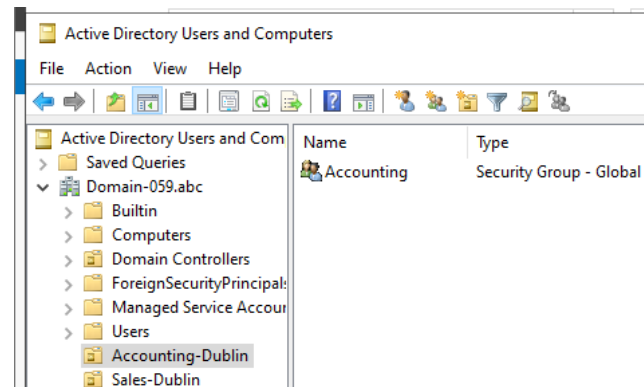
Part 2: Setting up the resources and network users

In this Second part I'll be setting up the Global Security groups inside the Organizational Units (Part 2.A). Then I'll add the new DigiTech users (Part 2.B) and each of them to their designated department (Part 2.C) and give their permissions to access files (Part 2.D) and find the files (Part 2.E). Finally, set password policies for the users (Part 2.F).

2.A Global Security Group

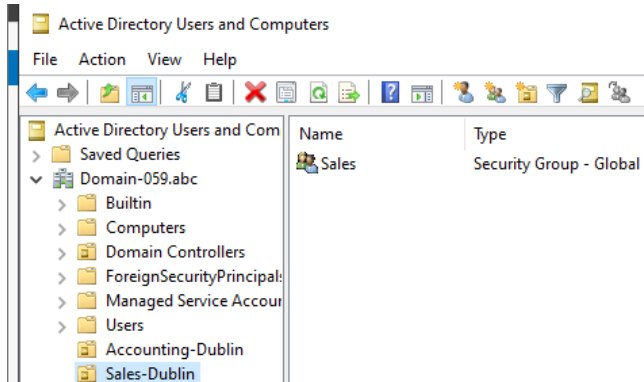


Accounting



12 Creating the Global Security Group called Accounting into the Accounting-Dublin OU.

Sales



13 creating the Global Security Group called Sales into the Sales-Dublin OU.

2.B Creating Users

Accounting Users

Creating the Accounting Users

14 All Accounting Users

15 Creating User in Accounting

Creating the Accounting users with their Name, User logon name and all of them must change the password at their next logon.

Name	Type	Description
Accounting	Security Group - Global	
Patrick J Jones	User	
Patrick L Jones	User	
Patrick Richardson	User	
Barb Dwyer	User	
Cherry Pies	User	

Sales Users

Creating the Sales Users

16 Creating User in Sales

Creating the Sales users with their Name, User logon name and all of them must change the password at their next logon.

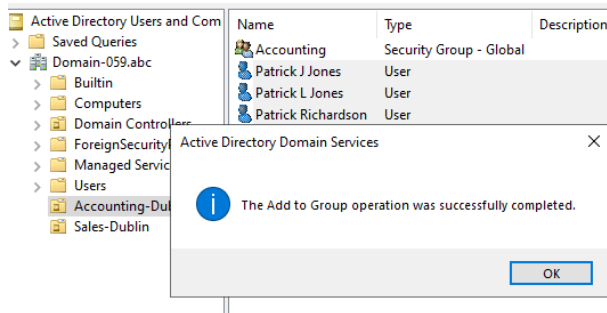
17 All users in Sales

Name	Type	Description
Claire Galway	User	
Claire Morris	User	
David R Doran	User	
Debra J Doran	User	
Justin Case	User	
Sales	Security Group - Global	

2.C Adding Users to Groups

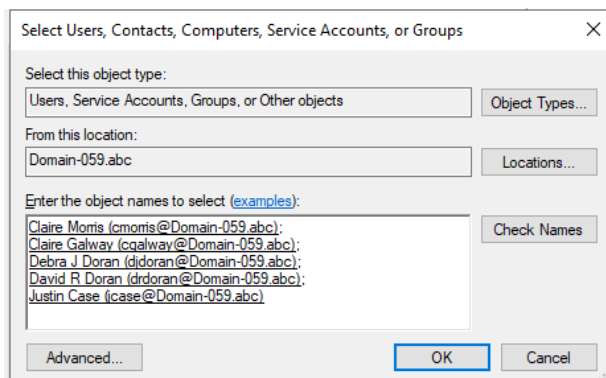
Using the Active Directory Users and Computers (ADUC) each of the users were added to their group.

19 Adding the new Accounting Users to the Accounting Group

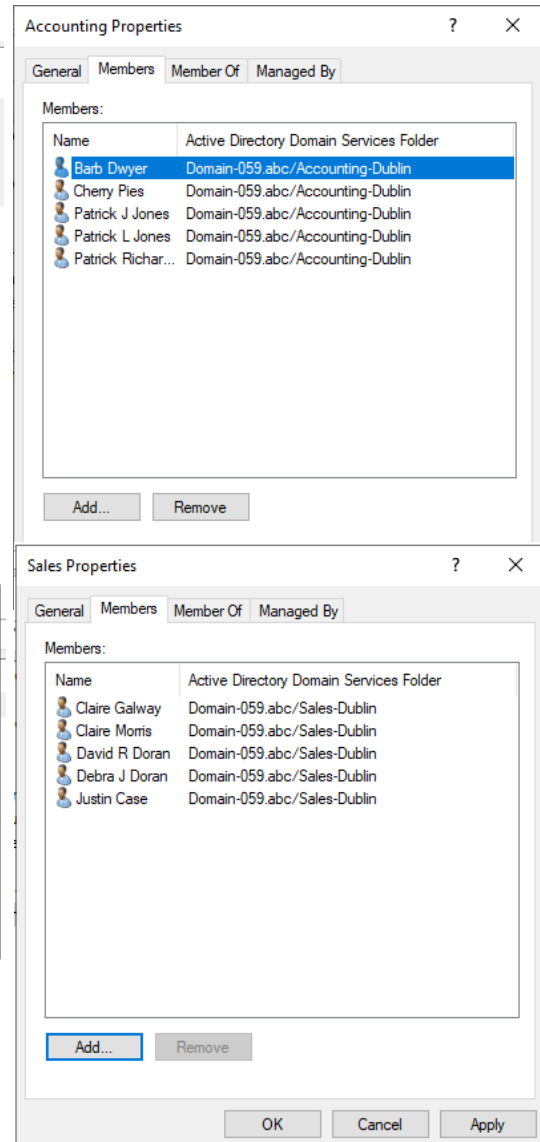


18 All member added to the Accounting Group

20 Adding the new Sales Users to the Sales Group

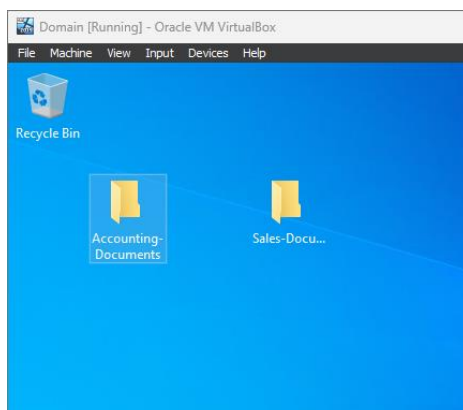


21 All members added to the Sales Group



2.C Creating Folder

22 The Folders for each Department

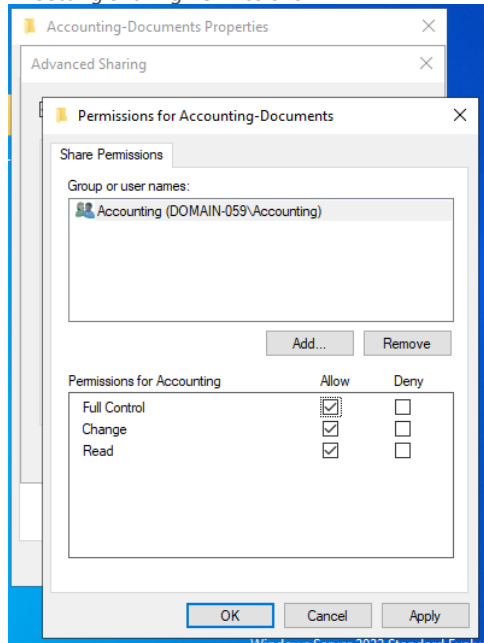


2.D Sharing Files

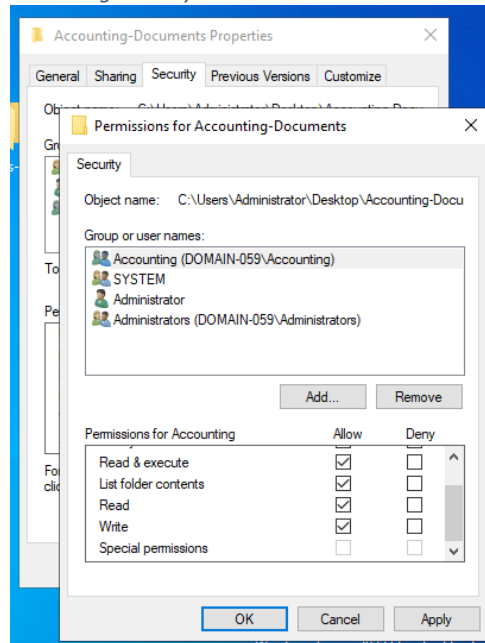
Accounting-Documents

Setting the folder permissions before sharing. Only the Accounting Group will be able to control this folder by giving them permission and removing the Everyone group.

24 Setting Sharing Permissions

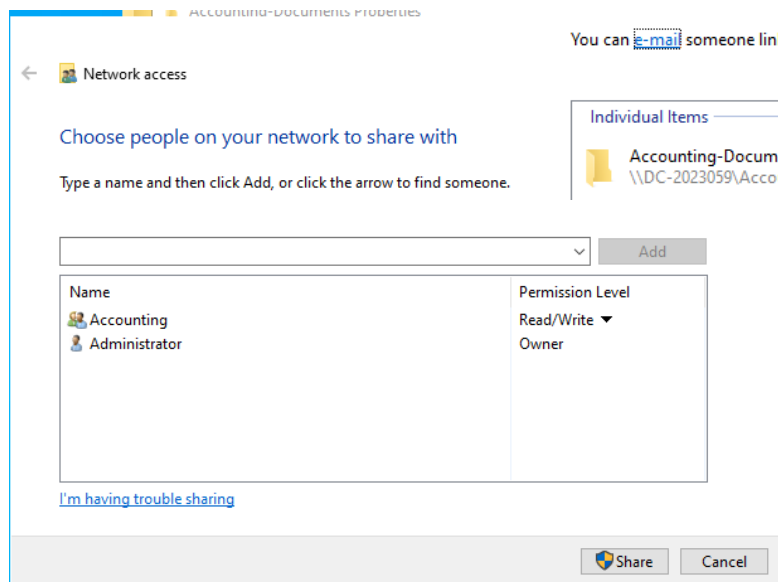


23 Setting Security Permission



Network access

26 Sharing the folder



Your folder is shared.

You can [e-mail](#) someone links to these shared items, or [copy](#) and paste the links into another app.

Individual Items

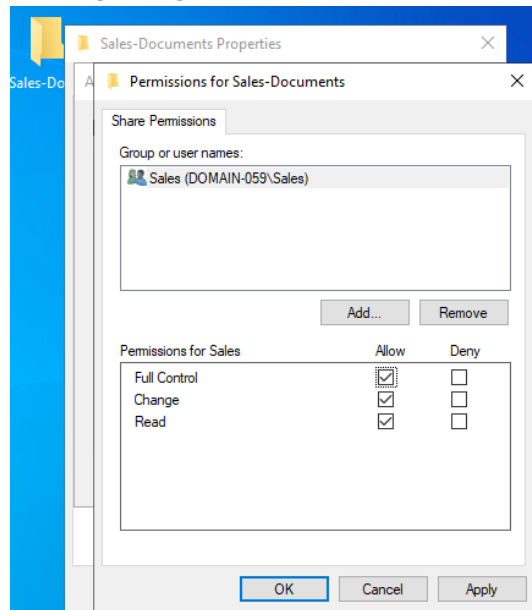
Accounting-Documents
\\DC-2023059\Accounting-Documents

25 Folder Shared and the UNC format to find it

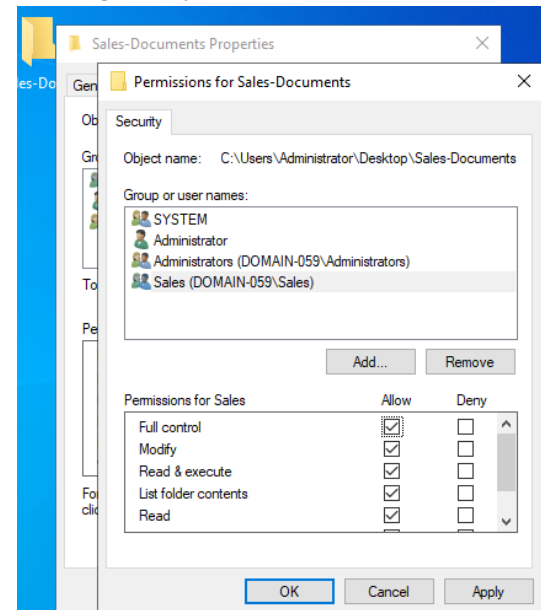
Sales-Documents

Setting the folder permissions before sharing. Only the Sales Group will be able to control this folder by giving them permission and removing the Everyone group.

29 Setting Sharing Permissions



28 Setting Security Permissions



27 Folder Shared and the UNC format to find it



Network access

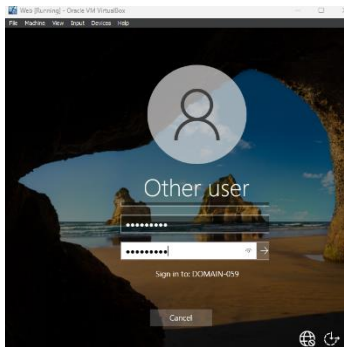
Your folder is shared.

You can [e-mail](#) someone links to these shared items, or [copy](#) and paste the links into another app.

Individual Items

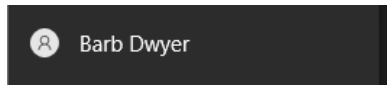
Sales-Documents
\\DC-2023059\Sales-Documents

2.E Checking the Permissions



30 Changing the User

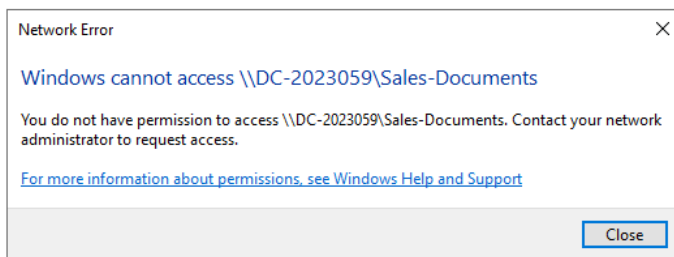
Barb Dwyer – Accounting User



Using the Barb Dwyer account to try to access both folders.

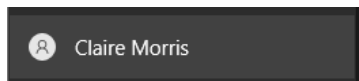
She could only access the Accounting folder.

32 Successfully accessing the accounting Folder



33 Failing on accessing the Sales folder

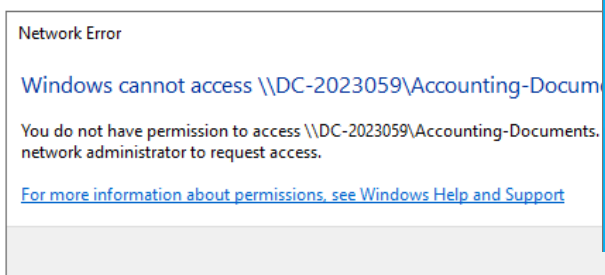
Claire Morris –Sales User



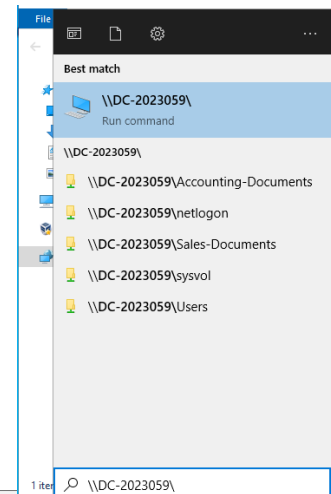
Using the Claire Morris account to try to access both folders.

She could only access the Sales folder.

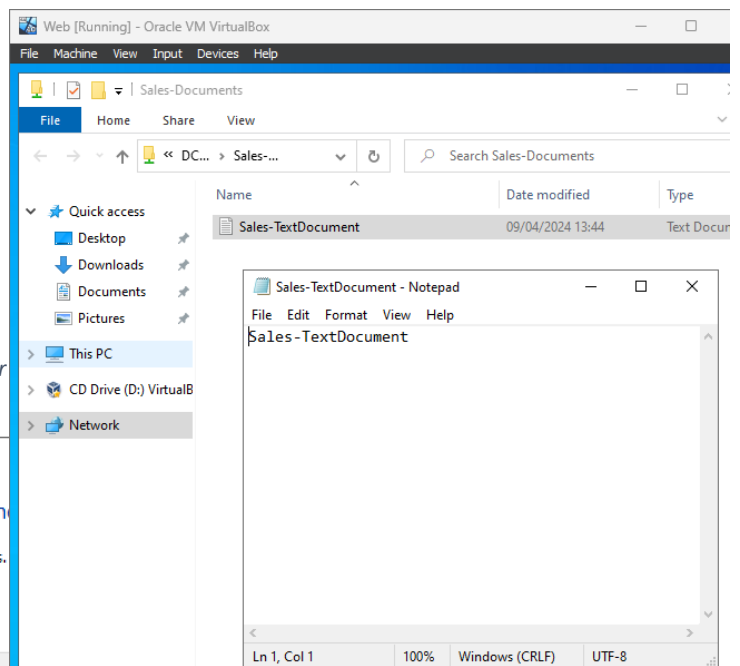
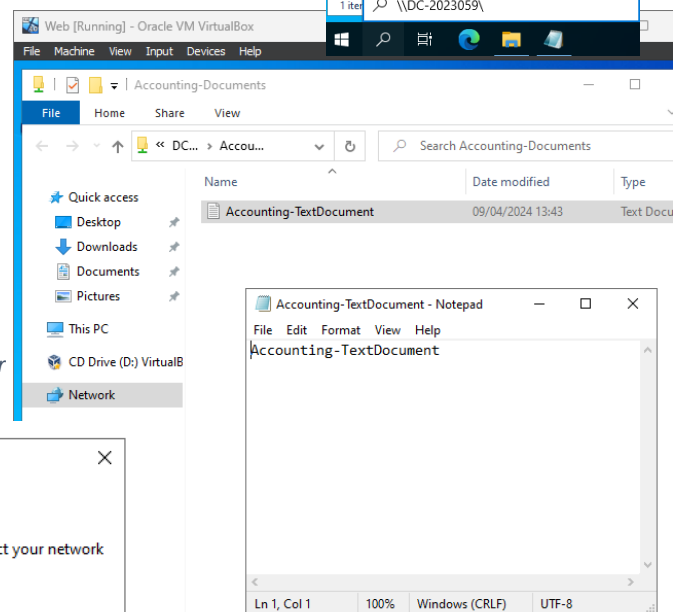
34 Successfully accessing the Sales Folder



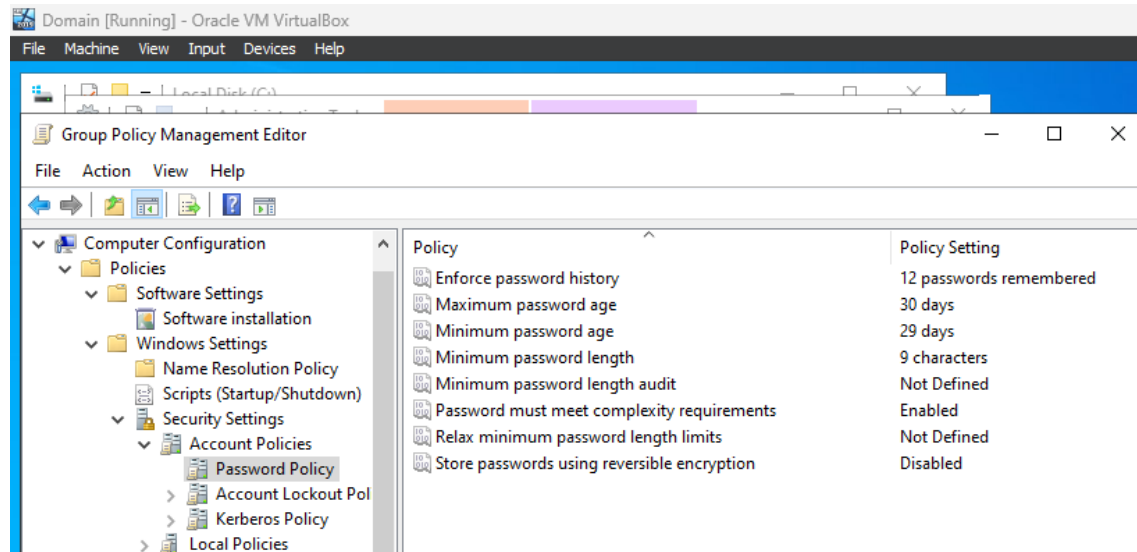
35 Failing on accessing the accounting folder



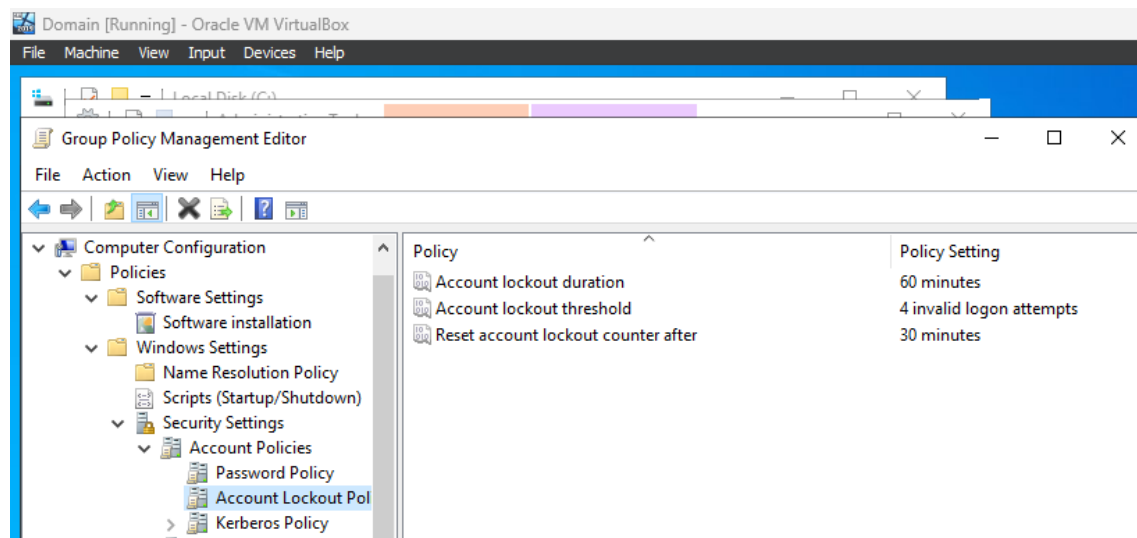
31 Searching through the UNC format



2.F Password and Account Lockout Policies



36 Setting the Password Policy for all users



37 Setting the Account Lockout Policy for all users

Part 3: Setting up and testing a Web Site

3.A Download IIS

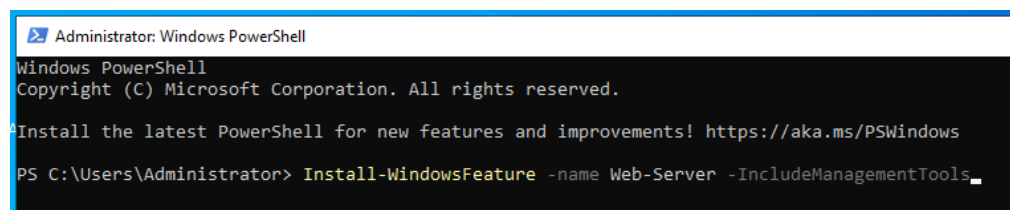


Figure 38 Installing the Web Server through PowerShell

3.B Web Page in the Web Server

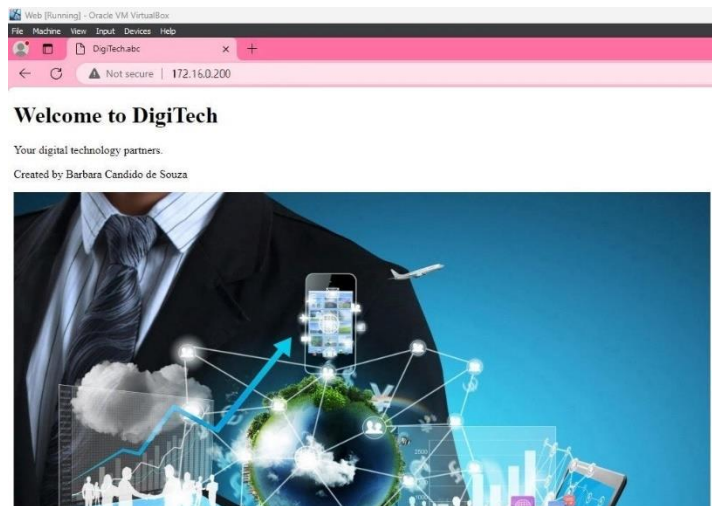


Figure 39 DigiTech Website on the Web Server

3.C Web Page in the Domain Server

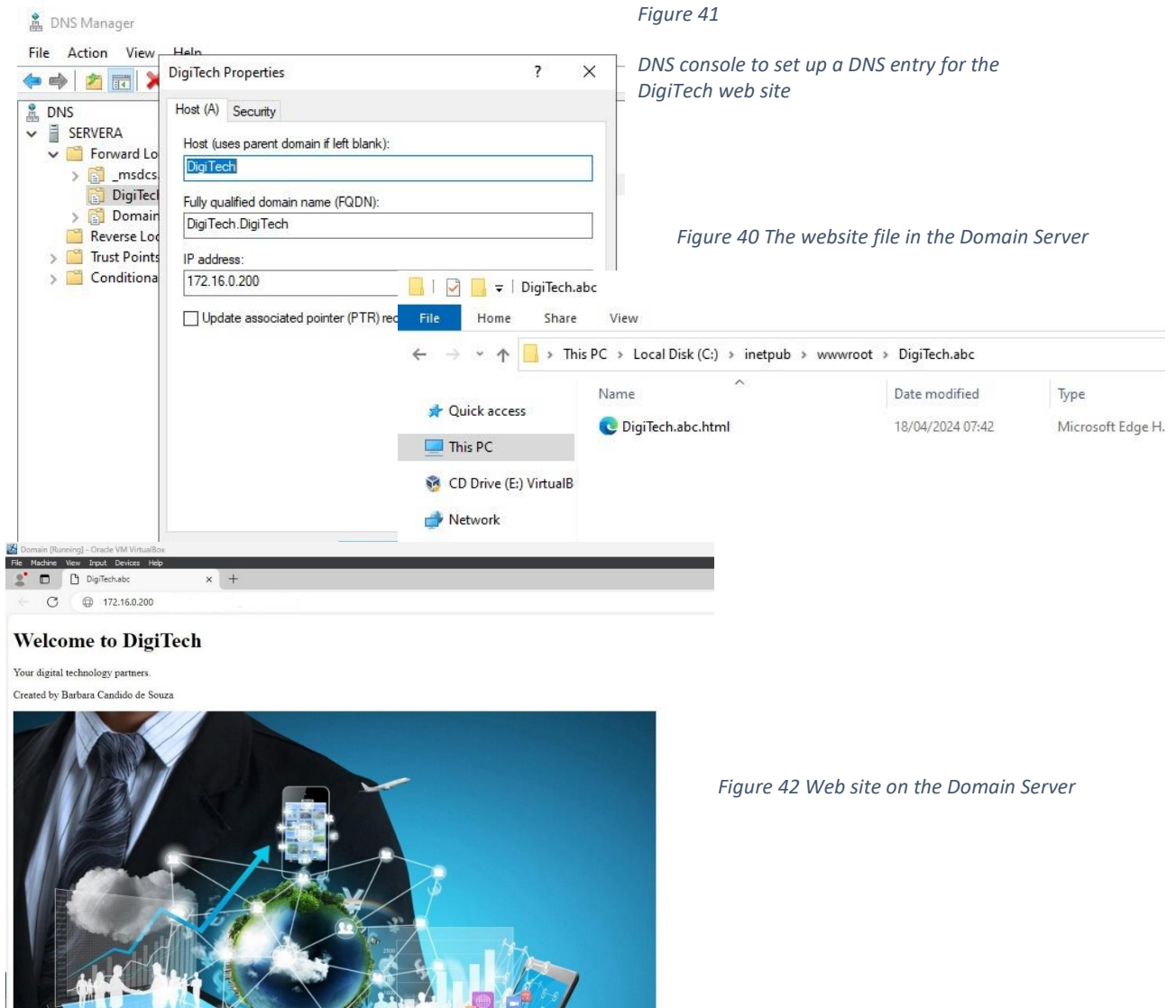


Figure 41

DNS console to set up a DNS entry for the DigiTech web site

Figure 40 The website file in the Domain Server

Figure 42 Web site on the Domain Server

Part 4: Setting Up a DHCP Server for the Domain

4.A Downloading DHCP

Add Roles and Features Wizard

Select server roles

Before You Begin
Installation Type
Server Selection
Server Roles
Features
DHCP Server
Confirmation
Results

Select one or more roles to install on the selected server.

Roles

- ☐ Active Directory Certificate Services
- ☒ Active Directory Domain Services (Installed)
- ☐ Active Directory Federation Services
- ☐ Active Directory Lightweight Directory Services
- ☐ Active Directory Rights Management Services
- ☐ Device Health Attestation
- ☒ **DHCP Server**
- ☒ DNS Server (Installed)
- ☐ Fax Server
- ☒ File and Storage Services (2 of 12 installed)
 - ☐ Host Guardian Service
 - ☐ Hyper-V
 - ☐ Network Policy and Access Services
 - ☐ Print and Document Services
 - ☐ Remote Access
 - ☐ Remote Desktop Services
 - ☐ Volume Activation Services
 - ☐ Web Server (IIS)
 - ☐ Windows Deployment Services
 - ☐ Windows Server Update Services

Describe the role

Dynai
Proto
to cel
provi
relate
comp

View installation progress

Starting installation

DHCP Server
Remote Server Administration Tools
Role Administration Tools
DHCP Server Tools

< Previous Next >

43 Adding the DHCP server role to the Domain Controller server computer using the Add Roles and Features in the Server Manager console

Server Manager

DHCP

File Action View Help

Dashboard DHCP Contents of DHCP Status Actions

Local Services dc-202305

All Servers

AD DS

DHCP

DNS

File and Storage

New Scope Wizard

Scope Name
You have to provide an identifying scope name. You also have the option of providing a description.

Type a name and description for this scope. This information helps you quickly identify how the scope is to be used on your network.

Name: Domain-059.abc

Description:

44 Configuring the DHCP using the DHCP console in Administrative Tools and adding the Scope name

New Scope Wizard

IP Address Range
You define the scope address range by identifying a set of consecutive IP addresses.

Configuration settings for DHCP Server

Enter the range of addresses that the scope distributes.

Start IP address: 172.16.0.1

End IP address: 172.16.0.254

Configuration settings that propagate to DHCP Client

Length: 16

Subnet mask: 255.255.0.0

< Back Next > Cancel

45 Setting the Starting and Ending IP Addresses and subnet mask

New Scope Wizard

Router (Default Gateway)
You can specify the routers, or default gateways, to be distributed by this scope.

To add an IP address for a router used by clients, enter the address below.

IP address: 172.16.0.1

Add Remove

46 Setting the Default Gateway to 172.16.0.1

New Scope Wizard

Lease Duration

The lease duration specifies how long a client can use an IP address from this scope.

Lease durations should typically be equal to the average time the computer is connected to the same physical network. For mobile networks that consist mainly of portable computers or dial-up clients, shorter lease durations can be useful. Likewise, for a stable network that consists mainly of desktop computers at fixed locations, longer lease durations are more appropriate.

Set the duration for scope leases when distributed by this server.

Limited to:

Days: Hours: Minutes:

New Scope Wizard

Domain Name and DNS Servers

The Domain Name System (DNS) maps and translates domain names used by clients on your network.

You can specify the parent domain you want the client computers on your network to use for DNS name resolution.

Parent domain:

To configure scope clients to use DNS servers on your network, enter the IP addresses for those servers.

Server name:

IP address:

Add

Resolve

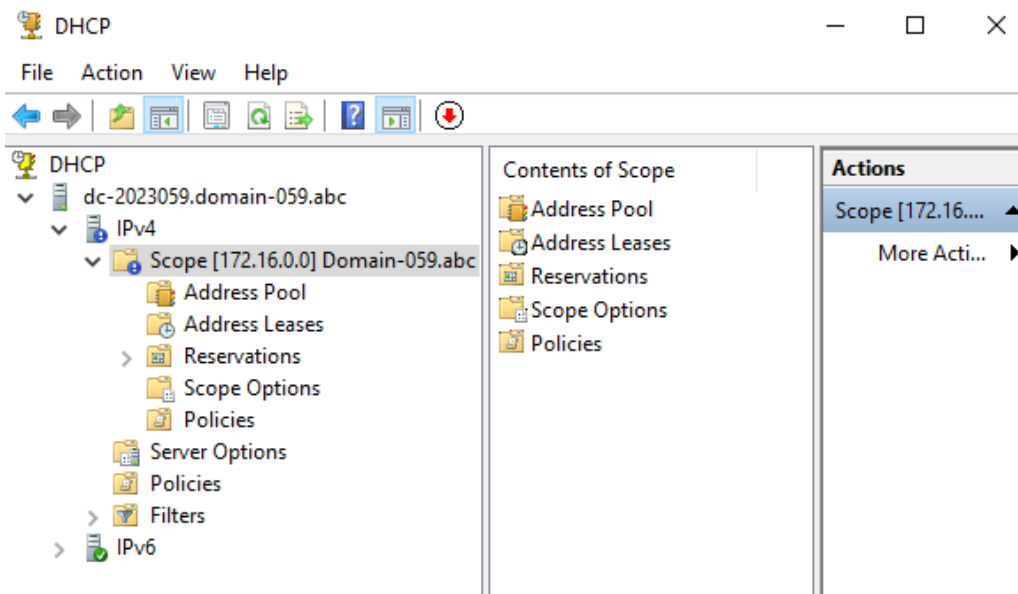
Remove

Up

Down

49 Setting the Lease Duration for 24 hours

48 DNS Server address 172.16.0.100



47 DHCP Ready

4.A Exclusions Range

New Scope Wizard

Add Exclusions and Delay

Exclusions are addresses or a range of addresses that are not distributed by the server. A delay is the time duration by which the server will delay the transmission of a DHCP OFFER message.

Type the IP address range that you want to exclude. If you want to exclude a single address, type an address in Start IP address only.

Start IP address: End IP address: Add

Excluded address range:

Address 172.16.0.1
Address 172.16.0.50
Address 172.16.0.100
Address 172.16.0.200

Remove

Subnet delay in milli second:

< Back

Next >

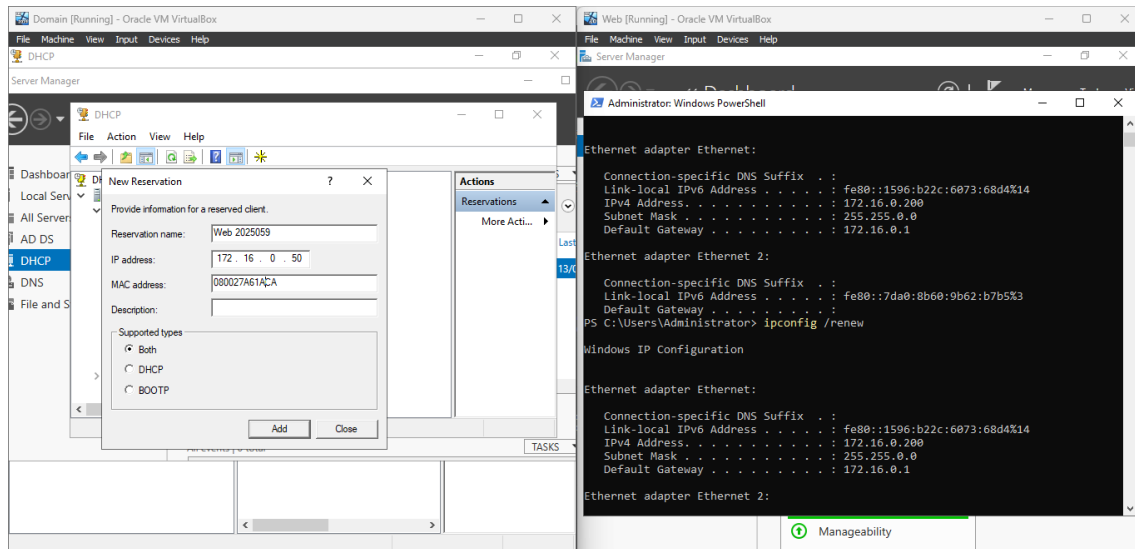
Cancel

	Start IP Address	End IP Address
DC DHCP		
dc-2023059.domain-059.abc		
IPv4		
Scope [172.16.0.0] Domain-059.abc		
Address Pool	172.16.0.1	172.16.0.254
Address Leases	172.16.0.1	172.16.0.1
Reservations	172.16.0.50	172.16.0.50
Scope Options	172.16.0.100	172.16.0.100
Policies	172.16.0.200	172.16.0.200
Server Options		
Policies		
Filters		
IPv6		

51 DHCP address exclusions ranges done

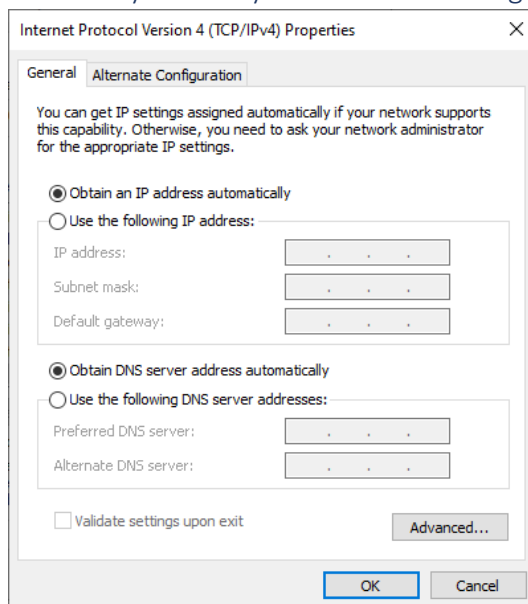
50 Adding DHCP address exclusions ranges

4.B Address Reservation



52 Creating an address reservation using the MAC Address

4.C Dynamically IP Address and Ping



54 Configuring the Web computer to dynamically receive the IP address

```
PS C:\Users\Administrator> ipconfig /renew

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix . : Domain-059.abc
    Link-local IPv6 Address . . . . . : fe80::1596:b22c:6073:68d4%14
    IPv4 Address. . . . . : 172.16.0.50
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 172.16.0.1
```

53 The computer getting the IP Reservation

```
Ethernet adapter Ethernet:

    Connection-specific DNS Suffix . : Domain-059.abc
    Link-local IPv6 Address . . . . . : fe80::1596:b22c:6073:68d4%14
    IPv4 Address. . . . . : 172.16.0.50
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 172.16.0.1

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix . : station
    Link-local IPv6 Address . . . . . : fe80::7da0:8b60:9b62:b7b5%13
    IPv4 Address. . . . . : 10.0.2.15
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.0.2.2

PS C:\Users\Administrator> ping 172.16.0.100

Pinging 172.16.0.100 with 32 bytes of data:
Reply from 172.16.0.100: bytes=32 time<1ms TTL=128
Reply from 172.16.0.100: bytes=32 time<1ms TTL=128
Reply from 172.16.0.100: bytes=32 time<1ms TTL=128
Reply from 172.16.0.100: bytes=32 time<1ms TTL=128

Ping statistics for 172.16.0.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
PS C:\Users\Administrator>
```

55 Pinging 172.16.0.100

```
Pinging 172.16.0.50 with 32 bytes of data:
Reply from 172.16.0.50: bytes=32 time<1ms TTL=128
Reply from 172.16.0.50: bytes=32 time<1ms TTL=128
Reply from 172.16.0.50: bytes=32 time<1ms TTL=128
Reply from 172.16.0.50: bytes=32 time<1ms TTL=128

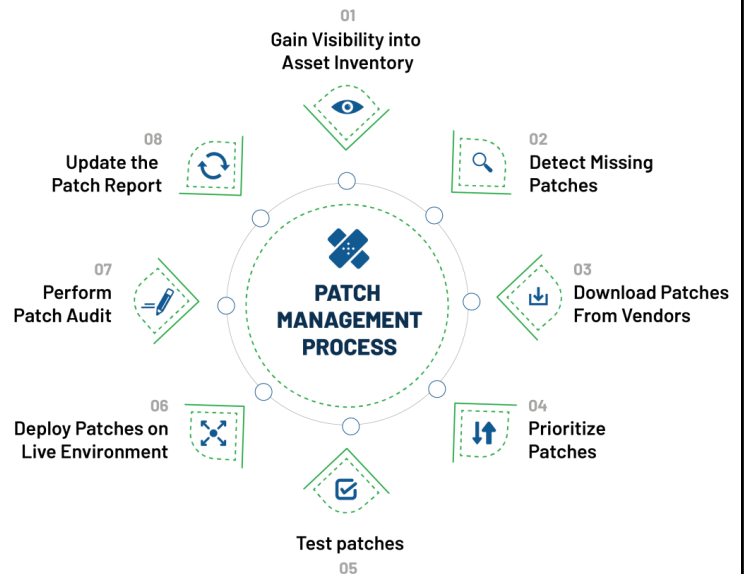
Ping statistics for 172.16.0.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
PS C:\Users\Administrator>
```

56 Pinging 172.16.0.50

Part 5: Patch Management and Update Servers for the Domain.

According to Essex (2024) , Patch Management is the process of identifying, testing, and applying patches to software systems. These patches are essentially quick fixes designed to enhance functionality and security, resolve issues and introduce new features (Posey, 2024).

Staying up to date with available patches is crucial, along with testing, to determine the specific patches required for each software and device. It is essential to ensure proper installation and to document the entire process for accountability, maintenance and future reference.



5.A Importance of patch management for network servers

Patch management is crucial for maintaining the security, reliability, and up-to-date functionality of computers and networks within organizations. It also plays a key role in ensuring compliance with security and privacy regulations. By applying patches, performance can be improved, and software can be kept compatible with the latest hardware.

Thangaraj (2022) outlines the significance of patch management by noting that all unpatched software poses high or critical risks, contributing to over 18% of network-level vulnerabilities.

Ester Shein (2022) highlights the risks associated with failed patch management, including security vulnerabilities, system downtime, errors, crashes, and data loss or corruption. Therefore, proper testing and execution are highly important

An example of a failed patch management that occurred within a large organization, was evidenced by a breach in the British Airways in 2018 (Airways, 2018) . They informed that data like personal and financial details of customers, were stolen. According to the BBC, after the incident they were fined £20m (Tidy, 2020). It's not clear on how the hackers were able to breach within the website, but BBC (2018) says they were able to "read between the lines" making it possible to attack. This incident shows the importance of effective patch management to try avoid such risks.

To prevent such occurrences, best practices for Patch Management include creating an inventory for the systems, determining risks and vulnerability, consolidating systems to the same version, testing patches before applying and apply them as quickly as possible, and finally, have a clear patch management policy. The benefits of following these practices are a secure environment, better user experience, latest innovation and it avoids penalties and fines.

Even though it's impossible to have a perfect patch management, according to EyeonTech (2022) organizations can significantly enhance IT system reliability by investing in a patch management policy, establishing a comprehensive patch management process, and using the appropriate software tools they will likely be successful at making their IT System more reliable, secure and current with the latest technology.

5.B Windows Server Update Services – WSUS

Computers typically install updates from Microsoft as soon as they are available, potentially causing issues such as different patch levels among client computers within an organization, software compatibility problems, or updates occurring during work hours leading to employee downtime. This is where Windows Server Update Services (WSUS) plays a crucial role.

As described by Wright (2024), WSUS is a Windows server role specifically designed to facilitate the planning, management, and deployment of updates, patches, and hotfixes for Windows servers, client operating systems, and other Microsoft software. WSUS services allows system administrators the opportunity to conduct thorough testing to ensure compatibility with their network infrastructure. It also provides administrators to control the timing and manner in which servers will receive updates, offering a centralized point for clients to obtain these updates.

In summary, WSUS facilitates effective update management by providing administrators with the tools to orchestrate update deployment, reduce operational risks associated with updates, and optimize the timing of installations to align with organizational needs and priorities.

Part 6: These are challenge tasks.

6. Downloading .msi Files

(How to Silently Install Adobe Acrobat Reader DC, 2021)

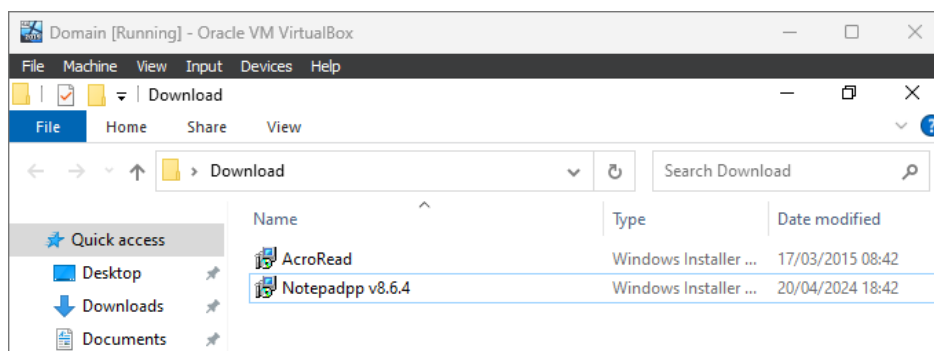


Figure 57 .msi files of Notepad and Adobe in the Shared Folder

6.A Group Policy Management - Sales

(Use Group Policy to remotely install software, 2024)

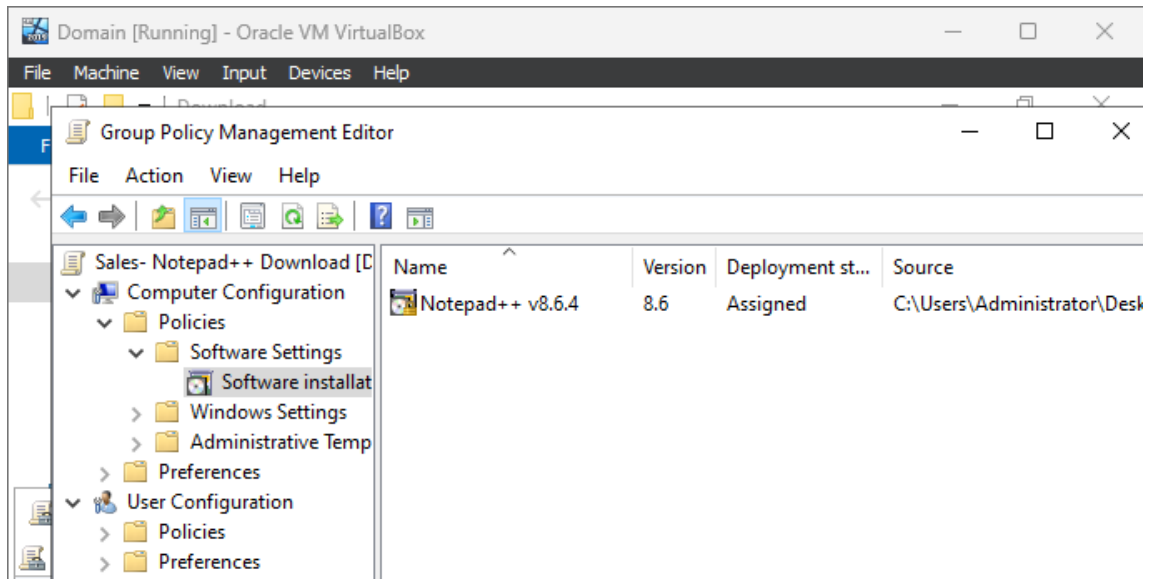


Figure 58 Adding a new GPO in Sales as a Software Installation

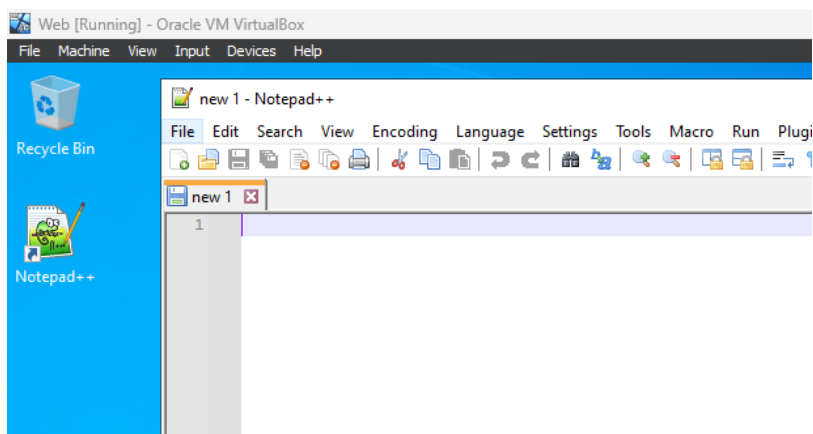


Figure 59 Sales User with the Notepad++

6.B Group Policy Management - Accounting

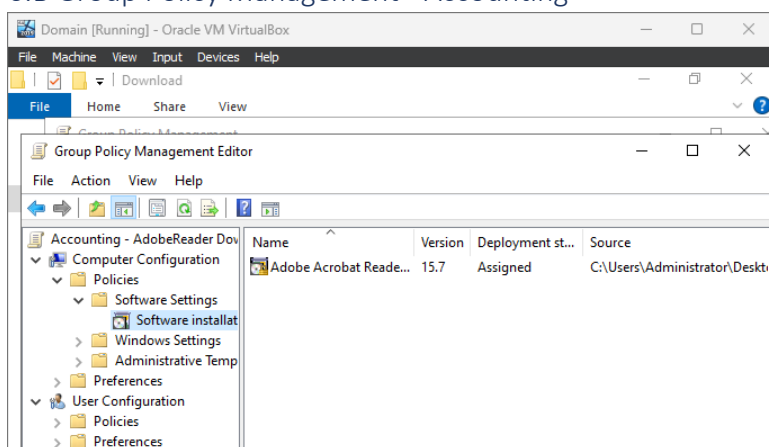


Figure 60 Adding a new GPO in Accounting as a Software Installation

6.C Wallpapers (Group Policy and Security with Windows Server, 2022)

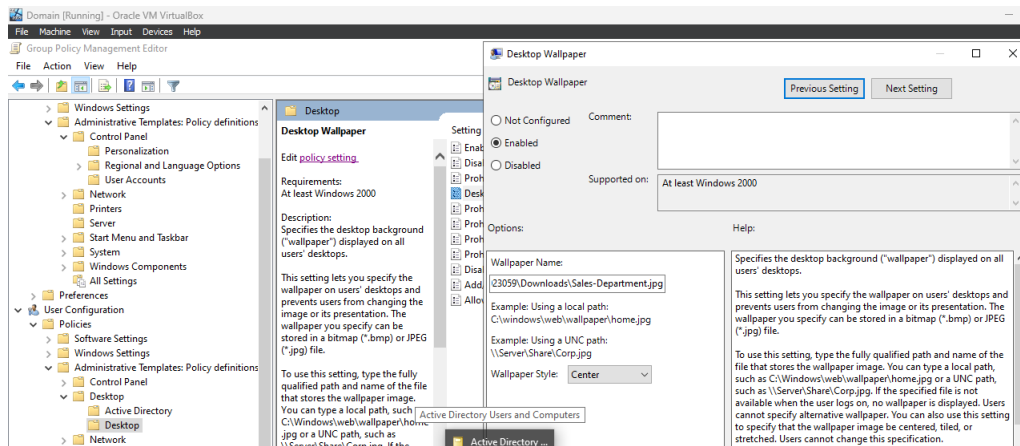


Figure 61 Enabling the Desktop Wallpaper policy and adding the new Wallpaper for Accounting

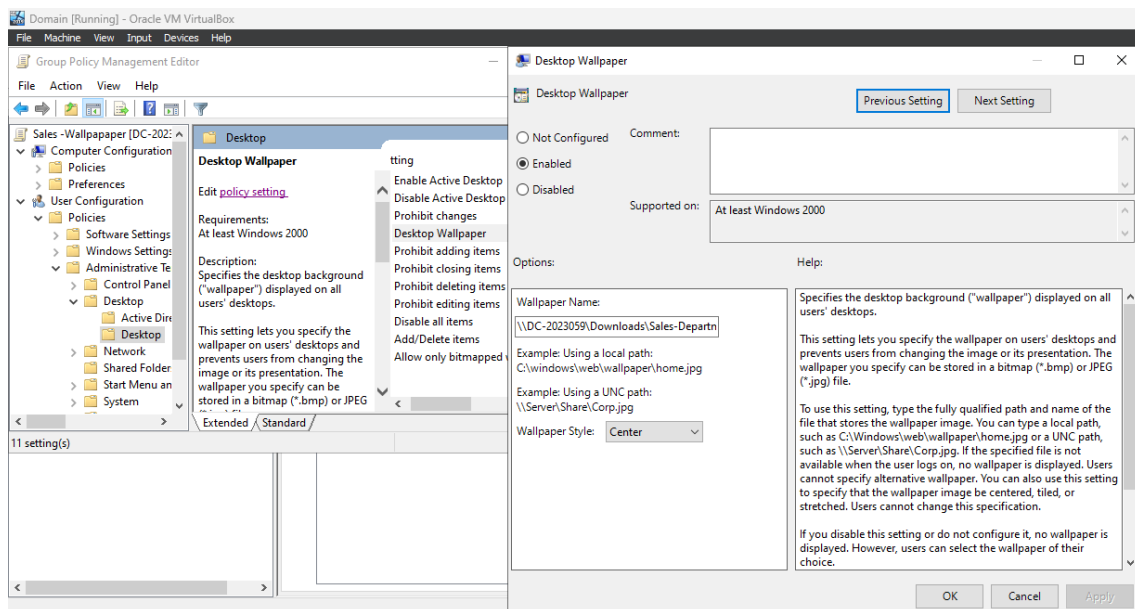


Figure 64 Enabling the Desktop Wallpaper policy and adding the new Wallpaper for Sales



Figure 63 New Wallpaper for Accounting



Figure 62 New Wallpaper for Sales

6.D Restrict Control Panel

(Restrict Control Panel Access using Group Policy, 2023)

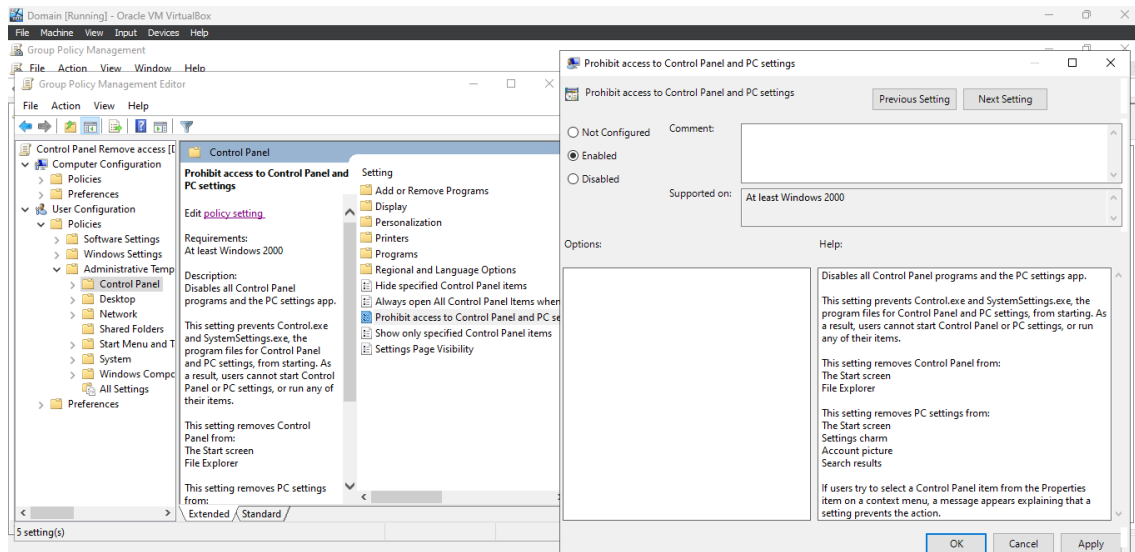


Figure 65 Prohibiting the Access to Control Panel in the Group Management Console for both Accounting and Sales

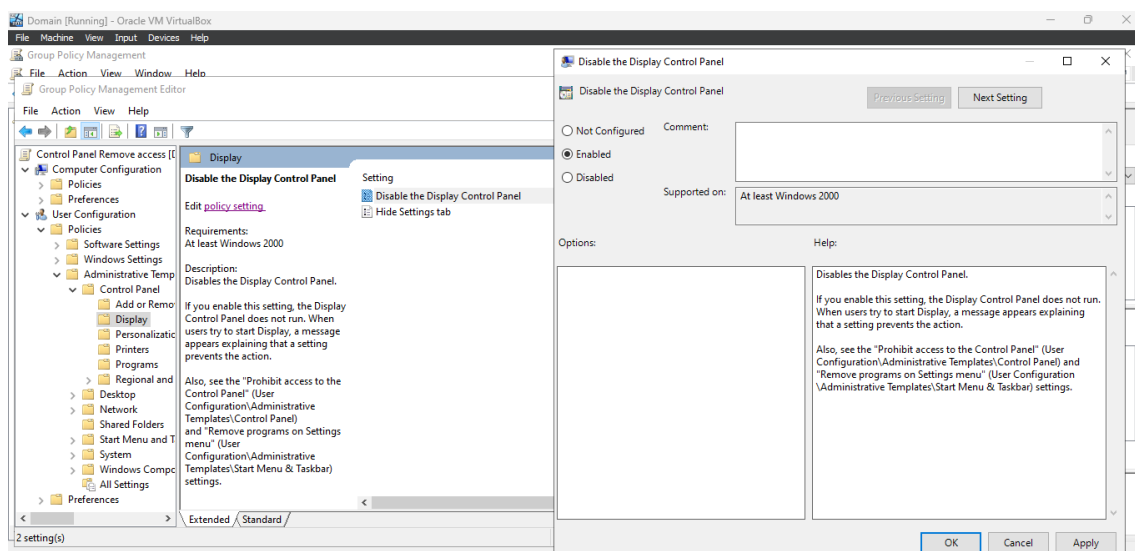


Figure 66 Disabling the Display of the Control Panel in the Group Management Console for both Accounting and Sales

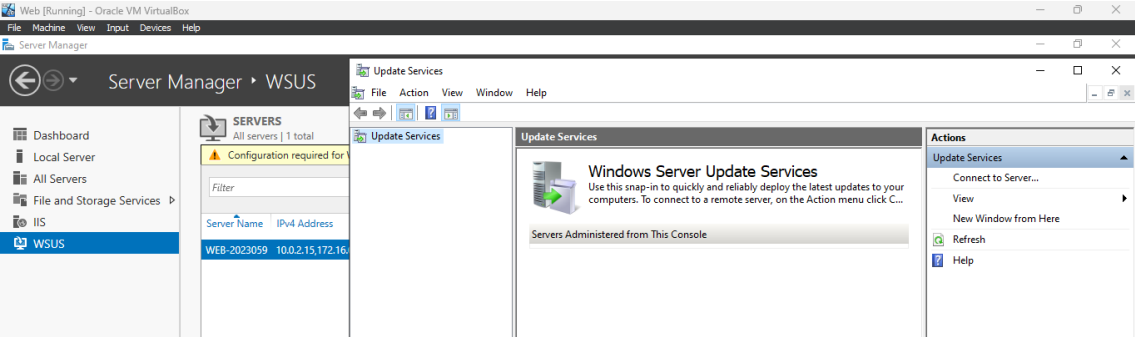
6.E

6.F Pricing and licensing for Windows Server 2022

Windows Server 2022 Edition	Ideal for	Licensing model	CAL requirements[1]	Suggested Retail Price (MSRP)[4]
Datacenter[2]	Highly virtualized datacenters and cloud environments	Core-based	Windows Server CAL	\$6,155
Standard[2]	Physical or minimally virtualized environments	Core-based	Windows Server CAL	\$1069
Essentials	Small businesses with up to 25 users and 50 devices	Specialty servers (server license)[3]	No CAL required	\$501

Figure 67 Price and License for Windows Server 2022 (Microsoft, 2024)

6.G WSUS



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